Supplementary Material for "Subgroup Identification and Variable Selection in the High-Dimensional Heterogeneous Cox Model"

S1 Some details of the algorithm

S1.1 Coordinate Descent Algorithm

For the fixed $\boldsymbol{\beta} = \widehat{\boldsymbol{\beta}}$, we consider the following objective function:

$$Q(\widehat{\boldsymbol{\beta}}, \boldsymbol{\eta}, \boldsymbol{\Xi}) = \widetilde{g}(\boldsymbol{\Xi}; \boldsymbol{\Xi}') + \sum_{i=1}^{q} P_{\lambda_2}^{(2)}(|\eta_i|).$$

According to [4], we consider the following weighted objective function to balance the regularization strengths on different components of $\boldsymbol{\eta}$:

$$\widetilde{Q}(\widehat{\boldsymbol{\beta}}, \boldsymbol{\eta}, \boldsymbol{\Xi}) = \widetilde{g}(\boldsymbol{\Xi}; \boldsymbol{\Xi}') + \sum_{i=1}^{q} \boldsymbol{G}_{ii} P_{\lambda_2}^{(2)}(|\eta_i|),$$

where $\boldsymbol{G} = \boldsymbol{Z}'\boldsymbol{H}\boldsymbol{Z}$ with $\boldsymbol{Z} = (\boldsymbol{Z}'(T_{ij}), j \in \mathcal{R}_i, i = 1, \ldots, n)$. It suffices to minimize the following objective function in order to update $\boldsymbol{\eta}$:

$$\overline{Q}(\widehat{\boldsymbol{\beta}}, \boldsymbol{\eta}, \boldsymbol{\Xi}) = \frac{1}{2}\boldsymbol{\eta}^\top \boldsymbol{H}\boldsymbol{\eta} - \boldsymbol{\eta}^\top \boldsymbol{b} + \sum_{i=1}^{q} \boldsymbol{G}_{ii} P_{\lambda_2}^{(2)}(|\eta_i|),$$

where $\boldsymbol{b} = \boldsymbol{Z}'\big(\boldsymbol{H}\boldsymbol{\Xi}' - \boldsymbol{H}\boldsymbol{X}\boldsymbol{\beta} - \bigtriangledown g(\boldsymbol{\Xi}')\big)$. The first order derivative at η_j can be estimated by solving

$$\frac{\partial \overline{Q}(\widehat{\boldsymbol{\beta}}, \boldsymbol{\eta}, \boldsymbol{\Xi})}{\partial \eta_j} = \boldsymbol{G}_{jj}\eta_j + \boldsymbol{\eta}_{-j}^\top \boldsymbol{G}_{j,-j} + \boldsymbol{G}_{j,j} P_{\lambda_2}^{(2)\prime}(|\eta_j|) = 0,$$

where $\boldsymbol{\eta}_{-i}$ represents the vector composing of the remaining $(i-1)$ elements of $\boldsymbol{\eta}$ after the ith element is removed, $\boldsymbol{G}_{i,-i}$ denotes the vector formed by the ith column of $\boldsymbol{G}$, with the ith element removed. Define $f_j = \big(b_j - \boldsymbol{\eta}_{-j}^\top \boldsymbol{G}_{j,-j}\big)\boldsymbol{G}_{j,-j}$, the coordinate descent algorithm for the group-SCAD penalty can be solved by using the univariate soft thresholding operator:

$$\eta_j = \begin{cases} \mathrm{ST}(f_j, \lambda_2), & |f_j| \le 2\lambda_2; \\ \frac{\mathrm{sign}(f_j)(|f_j| - \frac{\gamma_2\lambda_2}{\gamma_2 - 1})}{1 - 1/(\gamma_2 - 1)}, & 2\lambda_2 < |f_j| \le \gamma_2\lambda_2; \\ f_j, & |f_j| > \gamma_2\lambda_2, \end{cases} \tag{S1.1}$$

For the group-MCP penalty with parameter γ_2,

$$\eta_j = \begin{cases} \frac{\mathrm{ST}(f_j, \lambda_2)}{1 - 1/\gamma_2}, & |f_j| \le \gamma_2\lambda_2; \\ f_j, & |f_j| > \gamma_2\lambda_2, \end{cases} \tag{S1.2}$$

where $\mathrm{ST}(t, \lambda) = \mathrm{sign}(t)(|t| - \lambda)_+$ is the soft thresholding rule, and $(x)_+ = x$ if $x > 0$ and 0 otherwise. The Coordinate Descent Algorithm is summarized as follows.

Algorithm S.1. Coordinate Descent Algorithm

1. Set the initial value of $\widetilde{\boldsymbol{\eta}}$;

repeat

2. Compute $\widetilde{\boldsymbol{\Xi}}'$, $\bigtriangledown g(\widetilde{\boldsymbol{\Xi}}')$, and $\widetilde{\boldsymbol{b}}$;

3. For $j = 1, \ldots, q$, cyclically update the jth component $\widehat{\eta}_j$ of $\widehat{\boldsymbol{\eta}}$ using equations (S1.1) and (S1.2) for SCAD and MCP, respectively;

4. Set $\widetilde{\boldsymbol{\eta}} = \widehat{\boldsymbol{\eta}}$;

until convergence of $\widehat{\boldsymbol{\eta}}$.

S1.2 Majorized ADMM Algorithm

For the fixed $\boldsymbol{\eta} = \widehat{\boldsymbol{\eta}}$, let $\widehat{\mathcal{A}}$ is the active set of $\widehat{\eta}$. we only need to consider the objective function as follows:

$$Q(\boldsymbol{\beta}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}) = \widetilde{g}(\boldsymbol{\Xi}; \boldsymbol{\Xi}') + \sum_{1 \le i < j \le n} P_{\lambda_1}^{(1)}(\|\boldsymbol{\beta}_i - \boldsymbol{\beta}_j\|) \tag{S1.3}$$

Let $\boldsymbol{\delta}_{ij} = \boldsymbol{\beta}_i - \boldsymbol{\beta}_j$, we reformulate the optimal problem (**??**) as minimizing the following objective function:

$$\begin{aligned} &Q(\boldsymbol{\beta}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}, \boldsymbol{\delta}) = \widetilde{g}(\boldsymbol{\Xi}; \boldsymbol{\Xi}') + \sum_{1 \le i < j \le n} P_{\lambda_1}^{(1)}(\|\boldsymbol{\delta}_{ij}\|) \\ &\text{subject to } \boldsymbol{\beta}_i - \boldsymbol{\beta}_j - \boldsymbol{\delta}_{ij} = \mathbf{0}, \end{aligned} \tag{S1.4}$$

where $\boldsymbol{\delta} = \{\boldsymbol{\delta}_{i,j}^\top, i < j\}^\top$. Following [3], the augmented Lagrangian for (S1.4) is

$$\begin{aligned} Q'(\boldsymbol{\beta}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}, \boldsymbol{\delta}; \boldsymbol{\nu}, \boldsymbol{\varrho}, \boldsymbol{\Xi}') = \widetilde{g}(\boldsymbol{\Xi}; \boldsymbol{\Xi}') &+ \sum_{1 \le i < j \le n} P_{\lambda_1}^{(1)}(\|\boldsymbol{\delta}_{ij}\|) \\ &+ \sum_{i=1}^n \langle \nu_i, \Xi_i(T_i) - \boldsymbol{\beta}_{\boldsymbol{i}}^\top \boldsymbol{X}_i - \widehat{\boldsymbol{\eta}}^\top \boldsymbol{Z}_i(T_i) \rangle + \sum_{1 \le i < j \le n} \langle \boldsymbol{\varrho}_{ij}, \boldsymbol{\beta}_i - \boldsymbol{\beta}_j - \boldsymbol{\delta}_{ij} \rangle \\ &+ \frac{\rho}{2} \sum_{i=1}^n (\Xi_i(T_i) - \boldsymbol{\beta}_{\boldsymbol{i}}^\top \boldsymbol{X}_i - \widehat{\boldsymbol{\eta}}^\top \boldsymbol{Z}_i(T_i))^2 + \frac{\rho}{2} \sum_{1 \le i < j \le n} \|\boldsymbol{\varrho}_{ij}, \boldsymbol{\beta}_i - \boldsymbol{\beta}_j - \boldsymbol{\delta}_{ij}\|^2, \end{aligned} \tag{S1.5}$$

where the Lagrangian dual variables $\boldsymbol{\nu} = (\nu_1 \ldots, \nu_n)^\top$ and $\boldsymbol{\varrho} = (\boldsymbol{\varrho}_{ij}^\top, i < j)^\top$ is the penalty parameter. This optimal problem is equivalent to (S1.4).

At the mth iteration, for a given $(\boldsymbol{\beta}^{(m-1)}, \boldsymbol{\Xi}^{(m-1)(t)}, \boldsymbol{\delta}^{(m-1)}; \boldsymbol{\nu}^{(m-1)}, \boldsymbol{\varrho}^{(m-1)}, \boldsymbol{\Xi}'^{(m-1)})$, cluster size $K^{(m-1)}$ and subgroup set $\mathcal{G}^{(m-1)}$, we update the parameters by the following steps:

<u>Step 1.</u> Updata $\boldsymbol{\beta}^{(m)}$ by minimizing

$$Q'(\boldsymbol{\beta}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}^{(m-1)}, \boldsymbol{\delta}^{(m-1)}; \boldsymbol{\nu}^{(m-1)}, \boldsymbol{\varrho}^{(m-1)}, \boldsymbol{\Xi}'^{(m-1)}).$$

For a given $(\boldsymbol{\Xi}, \boldsymbol{\delta}; \boldsymbol{\nu}, \boldsymbol{\varrho}, \boldsymbol{\Xi})$, it is equivalent to the minimizing the following function:

$$\begin{aligned} &\sum_{i=1}^n \langle \nu_i, Y_i(t) - \boldsymbol{\beta}_{\boldsymbol{i}}^\top \boldsymbol{X}_i - \widehat{\boldsymbol{\eta}}^\top \boldsymbol{Z}_i(t) \rangle + \sum_{1 \le i < j \le n} \langle \boldsymbol{\varrho}_{ij}, \boldsymbol{\beta}_i - \boldsymbol{\beta}_j - \boldsymbol{\delta}_{ij} \rangle \\ &+ \frac{\rho}{2} \sum_{i=1}^n (Y_i(t) - \boldsymbol{\beta}_{\boldsymbol{i}}^\top \boldsymbol{X}_i - \widehat{\boldsymbol{\eta}}^\top \boldsymbol{Z}_i(t))^2 + \frac{\rho}{2} \sum_{1 \le i < j \le n} \|\boldsymbol{\varrho}_{ij}, \boldsymbol{\beta}_i - \boldsymbol{\beta}_j - \boldsymbol{\delta}_{ij}\|^2. \end{aligned}$$

This is also equivalent to minimizing

$$\begin{aligned}&\langle \boldsymbol{\nu}, \boldsymbol{\Xi} - \boldsymbol{Z}\widehat{\boldsymbol{\eta}} - \widetilde{\boldsymbol{X}}_{\mathcal{G}}\boldsymbol{\alpha}\rangle + \langle \boldsymbol{\varrho}, \widetilde{\boldsymbol{A}}\boldsymbol{\alpha} - \boldsymbol{\delta}\rangle \\ &+ \frac{\rho}{2}\|\boldsymbol{\Xi} - \boldsymbol{Z}\widehat{\boldsymbol{\eta}} - \widetilde{\boldsymbol{X}}_{\mathcal{G}}\boldsymbol{\alpha}\|^2 + \frac{\rho}{2}\|\widetilde{\boldsymbol{A}}\boldsymbol{\alpha} - \boldsymbol{\delta}\|^2,\end{aligned} \tag{S1.6}$$

where $\widetilde{\boldsymbol{A}} = \boldsymbol{A}\widetilde{\boldsymbol{W}}_{\mathcal{G}}$. By setting the first derivative of (S1.6) to equal to zero, we get

$$\begin{aligned}\boldsymbol{\alpha}^{(m)} = \big(\widetilde{\boldsymbol{A}}^\top\widetilde{\boldsymbol{A}} + \widetilde{\boldsymbol{X}}_{\mathcal{G}^{(m-1)}}^\top\widetilde{\boldsymbol{X}}_{\mathcal{G}^{(m-1)}}\big)^{-1}\big(\widetilde{\boldsymbol{X}}_{\mathcal{G}^{(m-1)}}^\top\big(\boldsymbol{\Xi} + \boldsymbol{\nu}/\rho - \boldsymbol{Z}\widehat{\boldsymbol{\eta}}\big) + \widetilde{\boldsymbol{A}}^\top\big(\boldsymbol{\delta} - \boldsymbol{\varrho}/\rho\big)\big), \\ \boldsymbol{\beta}^{(m)} = \widetilde{\boldsymbol{W}}_{\mathcal{G}^{(m-1)}}\boldsymbol{\alpha}^{(m)}.\end{aligned} \tag{S1.7}$$

<u>Step 2.</u> Update $\boldsymbol{\Xi}^{(m)}$ by minimizing

$$Q'(\boldsymbol{\beta}^{(m)}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}, \boldsymbol{\delta}^{(m-1)}; \boldsymbol{\nu}^{(m-1)}, \boldsymbol{\varrho}^{(m-1)}, \boldsymbol{\Xi}'^{(m-1)}).$$

It is equivalent to the minimizing the following function:

$$\begin{aligned}&\langle \boldsymbol{\Xi}, \bigtriangledown g(\boldsymbol{\Xi}')\rangle + \frac{1}{2}\|\boldsymbol{\Xi} - \boldsymbol{\Xi}'\|_{\boldsymbol{H}}^2 \\ &+ \sum_{i=1}^n \langle \nu_i, \Xi_i(T_j) - \boldsymbol{X}_i^\top\boldsymbol{\beta_i} - \boldsymbol{Z}_i'(T_j)\widehat{\boldsymbol{\eta}}\rangle + \frac{\rho}{2}\sum_{i=1}^n(\Xi_i(T_j) - \boldsymbol{\beta_i}^\top\boldsymbol{X}_i - \widehat{\boldsymbol{\eta}}^\top\boldsymbol{Z}_i(T_j))^2.\end{aligned} \tag{S1.8}$$

That is,

$$\Xi_i^{(m)}(T_j) = (\widetilde{h}_i + \rho)^{-1}\big[-\bigtriangledown_i g(\Xi'^{(m-1)}(T_j)) - \nu_i^{(m-1)} + \rho(\boldsymbol{X}_i^\top\boldsymbol{\beta_i}^{(m)} + \widehat{\boldsymbol{\eta}}^\top\boldsymbol{Z}_i(T_j)))\big]. \tag{S1.9}$$

Thus, $\Xi_i'^{(m)}(T_j)$ is updated by

$$\Xi_i'^{(m)}(T_j) = \boldsymbol{X}_i^\top\boldsymbol{\beta}_i + \widehat{\boldsymbol{\eta}}^\top\boldsymbol{Z}_i(T_j). \tag{S1.10}$$

<u>Step 3.</u> Update $\boldsymbol{\delta_{ij}}^{(m)}$ by minimizing

$$Q'(\boldsymbol{\beta}^{(m)}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}^{(m)}, \boldsymbol{\delta}; \boldsymbol{\nu}^{(m-1)}, \boldsymbol{\varrho}^{(m-1)}, \boldsymbol{\Xi}'^{(m)}).$$

For a given $(\boldsymbol{\beta}, \widehat{\boldsymbol{\eta}}, \boldsymbol{\Xi}, \boldsymbol{\nu}, \boldsymbol{\varrho}, \boldsymbol{\Xi}')$,

$$\boldsymbol{\delta}_{ij} = \arg\min_{\boldsymbol{\delta}_{ij}} \frac{1}{2}\|\boldsymbol{\beta}_i - \boldsymbol{\beta}_j + \frac{\boldsymbol{\varrho}_{ij}}{\rho} - \boldsymbol{\delta}_{ij}\|^2 + \frac{1}{\rho}P_{\lambda_1}^{(1)}(\|\boldsymbol{\delta_{ij}}\|). \tag{S1.11}$$

We can get the closed form of $\boldsymbol{\delta}_{ij}^{(m)}$. For the SCAD penalty with $\gamma_1 > 1/\rho + 1$, we have

$$\boldsymbol{\delta}_{ij}^{(m)} = \begin{cases}\mathrm{ST}(\boldsymbol{l}_{ij}^{(m-1)}, \lambda_1/\rho), & \|\boldsymbol{l}_{ij}^{(m-1)}\| \le \lambda_1 + \lambda_1/\rho; \\ \frac{\big(\rho(\gamma_1-1) - \gamma_1\lambda_1/\|\boldsymbol{l}_{ij}^{(m-1)}\|\big)\boldsymbol{l}_{ij}^{(m-1)}}{\rho\gamma_1 - \rho - 1}, & 2\lambda_1 + \lambda_1/\rho < \|\boldsymbol{l}_{ij}^{(m-1)}\| \le \gamma_1\lambda_1; \\ \boldsymbol{l}_{ij}^{(m-1)}, & \|\boldsymbol{l}_{ij}^{(m-1)}\| > \gamma_1\lambda_1,\end{cases} \tag{S1.12}$$

where $\boldsymbol{l}_{ij}^{(m-1)} = \boldsymbol{\beta}_i^{(m)} - \boldsymbol{\beta}_j^{(m)} + \frac{\boldsymbol{\delta}_{ij}^{(m-1)}}{\rho}$. For the MCP penalty with $\gamma_1 > 1/\rho$, we have

$$\boldsymbol{\delta}_{ij}^{(m)} = \begin{cases}\mathrm{ST}(\frac{\rho\boldsymbol{l}_{ij}^{(m-1)}}{\rho - 1/\gamma_1}, \frac{\lambda_1}{\rho - 1/\gamma_1}), & \|\boldsymbol{l}_{ij}^{(m-1)}\| \le \gamma_1\lambda_1; \\ \boldsymbol{l}_{ij}^{(m-1)}, & \|\boldsymbol{l}_{ij}^{(m-1)}\| > \gamma_1\lambda_1.\end{cases} \tag{S1.13}$$

Step 4. Update $\boldsymbol{\nu}^{(m)}$ and $\varrho^{(m)}$ by

$$\begin{aligned} \nu_i^{(m)} &= \nu_i^{(m-1)} + \mathfrak{e}\varrho(Y_i^{(m)} - \boldsymbol{X}_i^\top \boldsymbol{\beta}_i^{(m)} - \widehat{\boldsymbol{\eta}}^\top \boldsymbol{Z}_i(t)); \\ \varrho_{ij}^{(m)} &= \varrho_{ij}^{(m-1)} + \mathfrak{e}\varrho(\boldsymbol{\beta}_i^{(m)} - \boldsymbol{\beta}_j^{(m)} - \boldsymbol{\delta}_{ij}^{(m)}), \end{aligned} \tag{S1.14}$$

where the constant $\mathfrak{e} \in (0, (1+\sqrt{5})/2)$.

Step 5. Update $K^{(m)}$ and $\mathcal{G}^{(m)}$ by $\widetilde{\boldsymbol{\delta}}_{ij}^{(m)}$, where

$$\widetilde{\boldsymbol{\delta}}_{ij}^{(m)} = \arg\min_{\widetilde{\boldsymbol{\delta}}_{ij}} \frac{1}{2}\|\boldsymbol{\beta}_i^{(m)} - \boldsymbol{\beta}_j^{(m)} - \widetilde{\boldsymbol{\delta}}_{ij}\|^2 + P_{\lambda_1}^{(1)}(\|\widetilde{\boldsymbol{\delta}}_{ij}\|). \tag{S1.15}$$

Here $\widetilde{\boldsymbol{\delta}}_{ij} = \mathbf{0}$ means that individuals i and j are in the same subgroup and can be used to update $\mathcal{G}$ and K.

As suggested by [5], we set $K^{(0)} = \lfloor\sqrt{n}\rfloor$ to ensure that it is sufficiently large, where $\lfloor a \rfloor$ denotes the largest integer no greater than a. A cluster analysis method can then be applied to determine $\mathcal{G}^{(0)} = (\mathcal{G}_1^{(0)}, \ldots, \mathcal{G}_{K^{(0)}}^{(0)})$. Let $\boldsymbol{\Xi}^{(0)} = \boldsymbol{\Xi}'^{(0)} = \boldsymbol{\beta}^{(0)T}\boldsymbol{X} + \widehat{\boldsymbol{\eta}}^\top \boldsymbol{Z}$, $\boldsymbol{\delta}^{(0)} = \boldsymbol{A}\boldsymbol{\beta}^{(0)}$, $\boldsymbol{\nu}^{(0)} = 0$, and $\boldsymbol{\varrho}^{(0)} = 0$. Define $r^{(m)} = \|\boldsymbol{\beta}^{(m)} - \boldsymbol{\beta}^{(m-1)}\| + |K^{(m)} - K^{(m-1)}|$. The Majorized ADMM Algorithm is summarized as follows.

Algorithm S.2. Majorized ADMM Algorithm

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1. Set $m \leftarrow 0$, initialize $(\boldsymbol{\beta}^{(0)}, \boldsymbol{\Xi}^{(0)}, \boldsymbol{\delta}^{(0)}; \boldsymbol{\nu}^{(0)}, \boldsymbol{\varrho}^{(0)}, \boldsymbol{\Xi}'^{(0)}), K^{(0)}$and $\mathcal{G}^{(0)}$;

repeat

2. $m \leftarrow m+1$

3. Update $\boldsymbol{\beta}^{(m)}$ by (S1.2), ($\boldsymbol{\Xi}^{(m)}$, $\boldsymbol{\Xi}'^{(m)}$) by (S1.9) and (S1.10), and $\boldsymbol{\delta}^{(m)}$ by (S1.12) and (S1.13) for SCAD and MCP respectively;

4. Update $\widetilde{\boldsymbol{\delta}}_{ij}^{(m)}$ by (S1.15), and then update $K^{(m)}$ and $\mathcal{G}^m$ according to $\widetilde{\boldsymbol{\delta}}_{ij}^{(m)}$;

until convergence of $r^{(m)}$.

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